

QUAN PHAM

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[Homepage](#) | [Google Scholar](#) | [LinkedIn](#)

RESEARCH INTEREST

I am interested in self-evolving agents and their applications to scientific research. I aim to develop agentic systems that effectively support researchers and the broader academic community. My current focus is on scalable, hierarchical skill libraries that allow agents to retrieve and reuse skills across tasks. I study how agents can build, organize, and continually improve these libraries through experience and feedback.

EDUCATION

National University of Singapore

PhD in Computer Science

Research Topic: Self-Evolving Agents, Reinforcement Learning

Advisor: Prof. Lee Wee Sun

Singapore

Aug, 2026 – Present

Hanoi University of Science and Technology

B.S in Data Science (GPA: 3.62/ 4.0)

Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Applied Statistics, Discrete Math, Math Modelling, Parallel Computing

Hanoi, Vietnam

Sep, 2021 – Mar, 2026

PUBLICATIONS

- Harit Vishwakarma, ..., **Quan Pham**, ..., Wee Sun Lee, Yee Whye Teh (2026). *Paperena: Reliable Science from Unreliable AI*. In *8th AI for Science Workshop (AI4Science @ NeurIPS 2026)*
- Dao Tran, Duc Anh Le, Ngoc Luu, **Quan Pham**, Tung Pham, Hung Bui (2026). *Beyond the Frontier: Stochastic Backtracking for Efficient Test-Time Scaling*. In *3rd AI for Math Workshop (AI4Math @ ICML 2026)*
- **Ngoc-Quan Pham**, Tuan Truong, Quyen Tran, Tan Nguyen, Dinh Phung, Trung Le (2025). *Promoting Ensemble Diversity with Interactive Bayesian Distributional Robustness for Fine-tuning Foundation Models*. In *International Conference on Machine Learning 2025 (ICML 2025)*
- Tuan Truong, Quyen Tran, **Ngoc-Quan Pham**, Dinh Phung, Nhat Ho, Trung Le (2025). *Improving Generalization with Flat Hilbert Bayesian Inference*. In *International Conference on Machine Learning 2025 (ICML 2025)*
- Pham Viet Thanh, Ngo Thi Thu Huyen, **Pham Ngoc Quan**, Nguyen Thi Thu Trang (2024). *A Robust Pitch-Fusion Model for Speech Emotion Recognition in Tonal Languages*. In *IEEE International Conference on Acoustics, Speech and Signal Processing 2024 (ICASSP 2024)*

RESEARCH EXPERIENCE

PhD Student, National University of Singapore

Advisor: Professor Lee Wee Sun

Aug, 2026 – Present

- Developed Paperena's auto-reviewer, combining structured PDF parsing with a 10-stage workflow for mapping claims to evidence, assessing novelty and rigor, and generating reviews through drafting and self-critique.
- Investigate hierarchical skill libraries for self-evolving agents, focusing on skill acquisition, retrieval, reuse, and refinement through experience and feedback.
- Co-develop the Reinforcement Learning Network framework, modeling neural computation as a Markov decision process to investigate input-conditioned network adaptation, uncertainty estimation, and continual learning.

AI Resident, Qualcomm AI Research

Mentor: Dr. Tung Pham, Dr. Hung Bui

Apr, 2025 – Jul, 2026

- Developed a sentence-level Sequential Monte Carlo framework for LLM mathematical reasoning, using process reward models to guide resampling and persistent sampling to preserve diverse reasoning trajectories.
- Investigated overconfidence and unreliability in process reward models to inform more robust search and backtracking strategies for test-time scaling.

- Contributed to two U.S. patent applications on efficient test-time scaling and backtracking-based reasoning, both currently under review.

AI Resident, VinAI Research

Aug, 2024 – Mar, 2025

Mentor: Prof. Trung Le, Prof. Dinh Phung

- Developed Interactive Bayesian Distributional Robustness to address ensemble collapse in parameter-efficient fine-tuning, modeling adapters as interacting Bayesian particles; first-author paper accepted at ICML 2025.
- Designed a diversity regularizer inspired by determinantal point processes and integrated it with distributionally robust optimization to encourage complementary ensemble predictions.
- Validated the proposed framework on VTAB-1K dataset with Vision Transformers and commonsense reasoning tasks with LLaMA-2, improving accuracy and calibration.

Research Assistant, Human Speech and Text Processing Lab

Sep, 2022 – May, 2024

Mentor: Dr. Nguyen Thi Thu Trang

- Implemented the data pipeline for ViSEC, a publicly available Vietnamese speech emotion recognition dataset.
- Co-developed a pitch-fusion model combining wav2vec 2.0 speech representations and pitch embeddings through cross-attention for emotion recognition in tonal languages; published at ICASSP 2024.
- Explored efficient attention mechanisms for efficient speech emotion recognition and proposed joint learning of categorical emotion labels and continuous arousal-valence-dominance representations to reduce sensitivity to label noise.

HONORS AND AWARDS

Research Scholarship, National University of Singapore	2026
ICML 2026 Silver Reviewer, International Conference on Machine Learning	2026
Student of 5 Merits, Hanoi University of Science and Technology	2026
First Prize in Vietnam National Startup Contest for Students (SV_STARTUP), Ministry of Education and Training	2023
Second Prize in SOICT Hackathon (AI-Powered BFIS track), Hanoi University of Science and Technology	2023
Third Prize in Vietnam Mathematical Olympiad, Ministry of Education and Training	2021

PROFESSIONAL SERVICE

Reviewer: ICML (26), AI4Math@ICML (26), MATH-AI@NeurIPS (26)

TEACHING EXPERIENCE

Introduction to Machine Learning, Maths and Science Summer School (MASSP)	2024
Functional Equation for Vietnam Mathematical Olympiad, Thai Binh High School for Gifted Students	2022

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, CUDA, SQL, Bash
Frameworks: PyTorch, Transformers, PEFT, vLLM, verl, SGLang, DeepSpeed, MCP
Tools: Git, Docker, Slurm, Weights & Biases

LANGUAGE

Vietnamese (Native), English (Advanced)